THE EVERYDAY CITY

A COMPLETE TOWN / SHARED GROUND / PRODUCTIVE EDGES

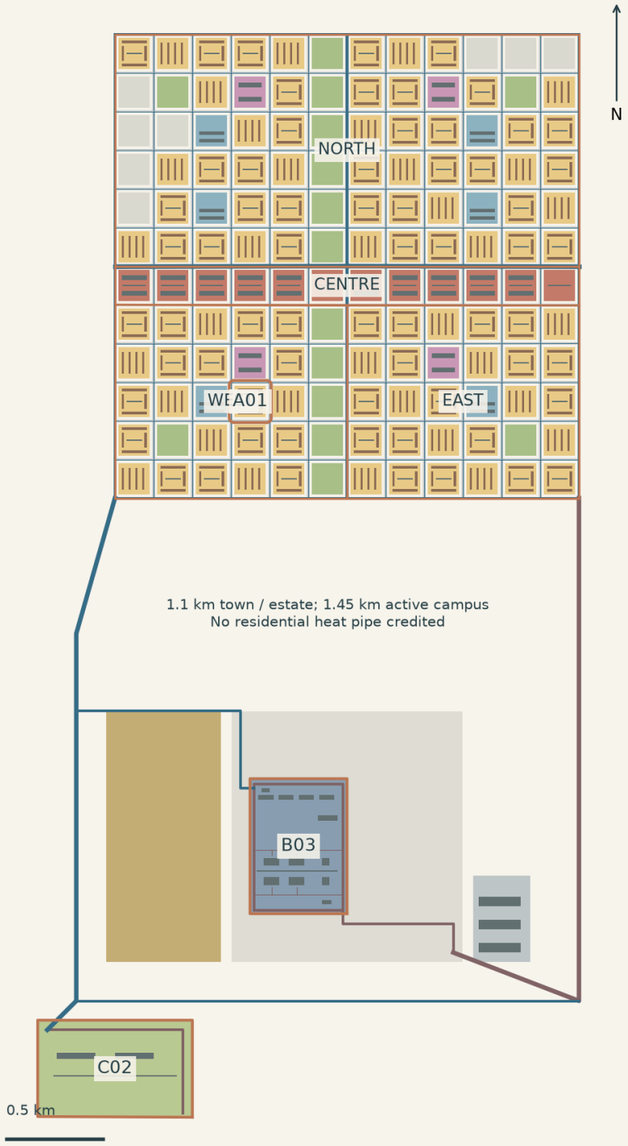


C02 / PROPOSED AGRIVOLTAIC LEARNING EDGE / LOCAL AERIAL

A continuous town of homes, schools and daily services. Production supports the city from its own campus. The cover shows the proposed farm-learning edge; the overall plan locates it within the productive landscape and town.

Make ordinary life the organising structure

The complete settlement is designed before the three detailed places. Buildings, roads, capacities, stages and camera controls share the same coordinates.



01 / A town with identifiable districts

West, East and North living districts form a continuous 2.4 x 2.4 km town. A civic band and dry commons connect them; housing is not scattered around compute halls.

02 / Production with a civic boundary

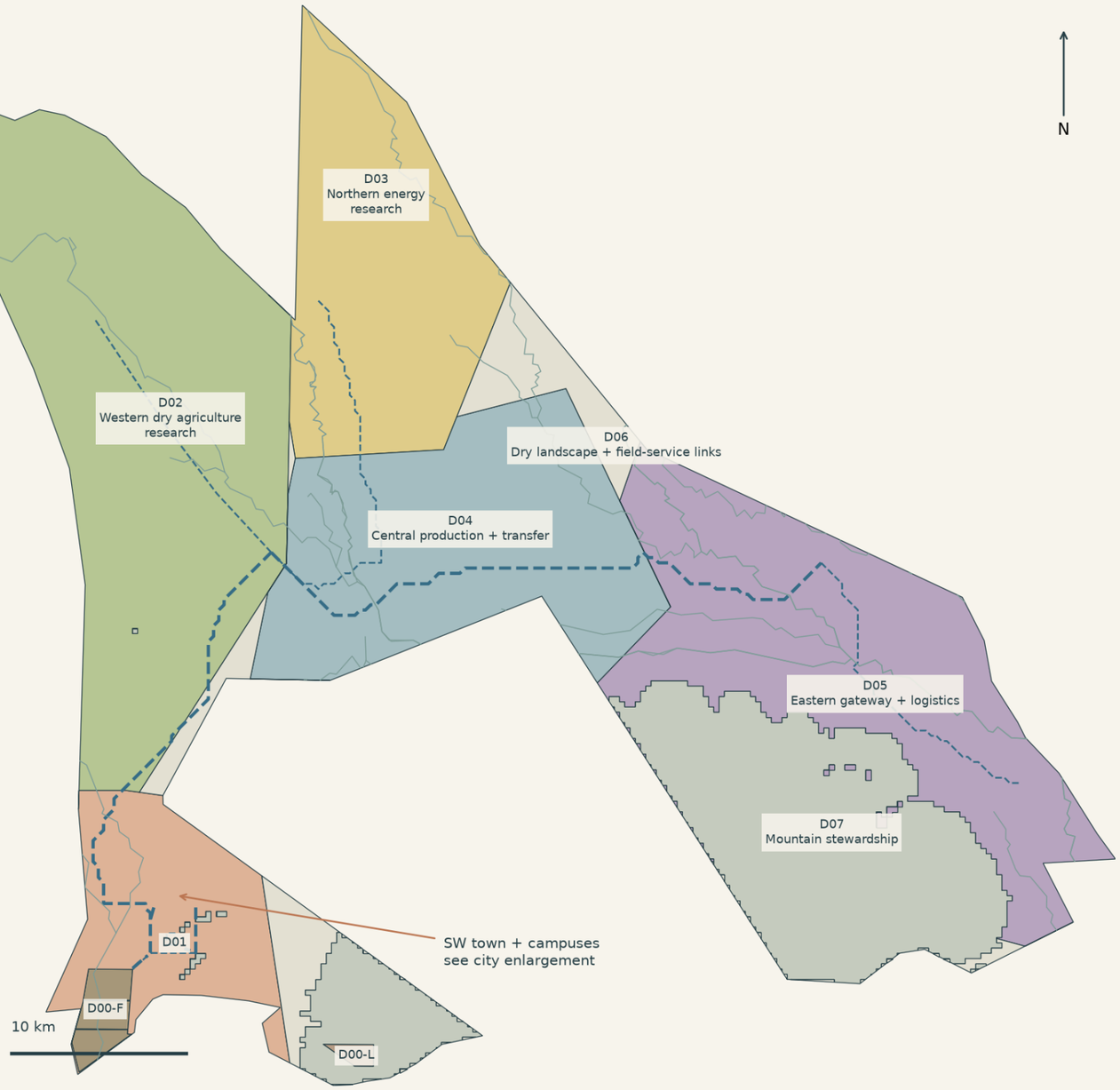
The productive estate sits 1.1 km south of town; its serviced campus is 1.45 km from town. Public arrival and production servicing take different approaches. No residential heat network is assumed.

03 / A useful working landscape

A new farm-learning interface is outside both fixed pilots. New trial rows can test the interface without claiming new irrigation rights or modifying the existing fields.

The whole issued site has a named role

Base zoning is mutually exclusive. Gross district area is not serviced construction, installed PV, irrigated agriculture or approved development.



D00-F / Fixed agrivoltaic pilot

13.9 km2 gross / design or fixed context

D00-L / Fixed living pilot

2.5 km2 gross / design or fixed context

D07 / Mountain stewardship

269.7 km2 gross / design or fixed context

D01 / Southwest settlement and farm services

113.2 km2 gross / design or fixed context

D02 / Western dry agriculture research

424.3 km2 gross / design or fixed context

D03 / Northern energy research

213.0 km2 gross / design or fixed context

D04 / Central production and transfer

251.9 km2 gross / design or fixed context

D05 / Eastern gateway and logistics

292.0 km2 gross / design or fixed context

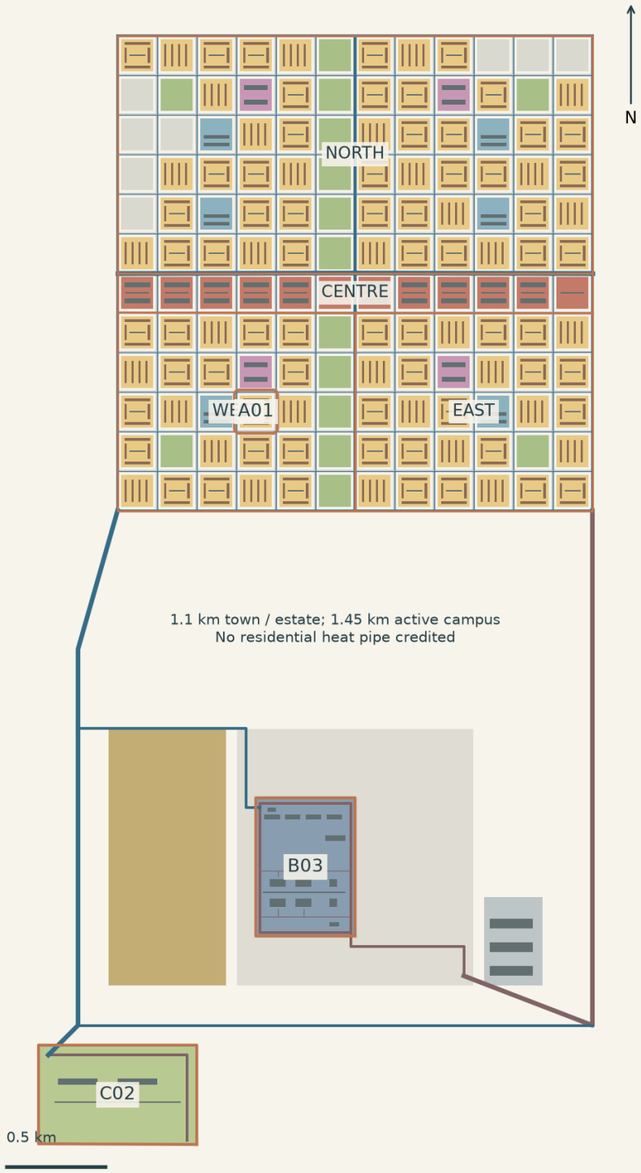
D06 / Dry landscape and field-service links

77.2 km2 gross / design or fixed context

Dashed blue routes are preliminary regional corridor studies. Thin green lines are source-derived seasonal channels. Neither implies a surveyed road right or permanent water.

Choose a coherent town, then accept its distances

The earlier compact all-functions rectangle is withdrawn. The new scheme keeps the southwest access basin while giving housing a complete and independently serviced urban footprint.



The selected urban ground

Town: 5.76 km2; median screened slope 1.18%, 95th percentile 4.25%. Three housing cells touching the preliminary flood screen are held without capacity.

The cost of an independent campus

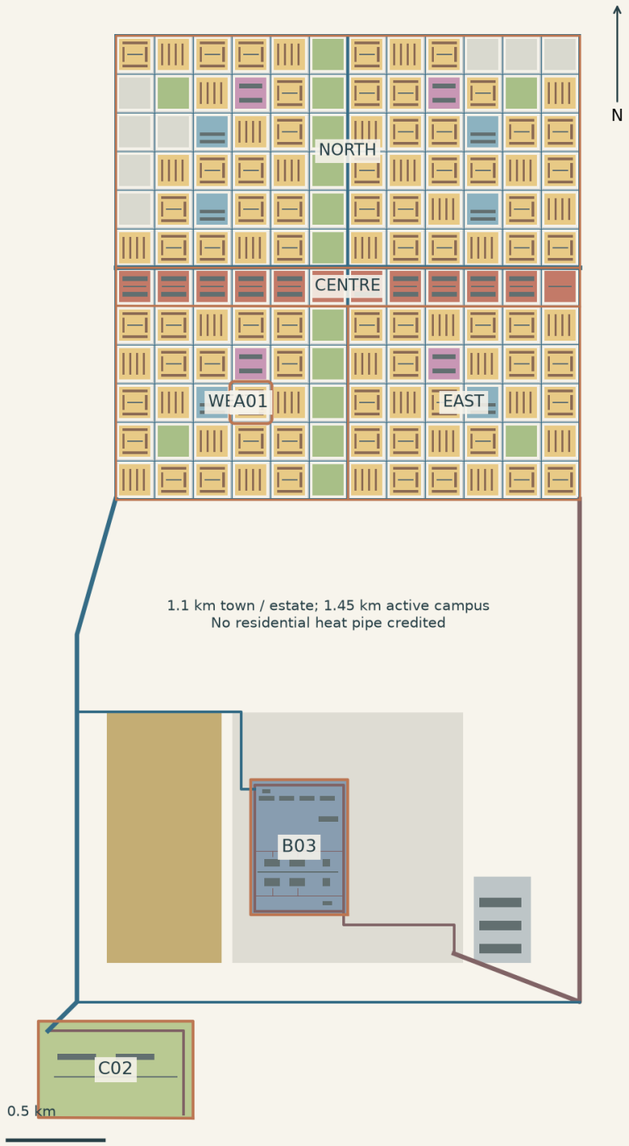
A compact co-location could shorten utilities and staff trips. The selected 1.1 km edge separation protects coherent residential streets and permits an operating boundary, but adds bus, utility and maintenance obligations.

What could reverse this decision

Geotechnical conditions, a lawful access corridor, measured noise/dust, utility connection costs or actual demand may change the town or campus boundary. The 150 m channel test is a declared screening distance, not a statutory setback.

One settlement system, several operating territories

City scale / true relative coordinates / all phases. The study footprint totals 8.635 km2; it is distinct from the full competition territory.



TOWN / residents and public life

Homes, shops, care and schools are organised by living district. The civic band is a shared destination; the north-south dry commons is public ground, not a lake.

WORKS / production and utility services

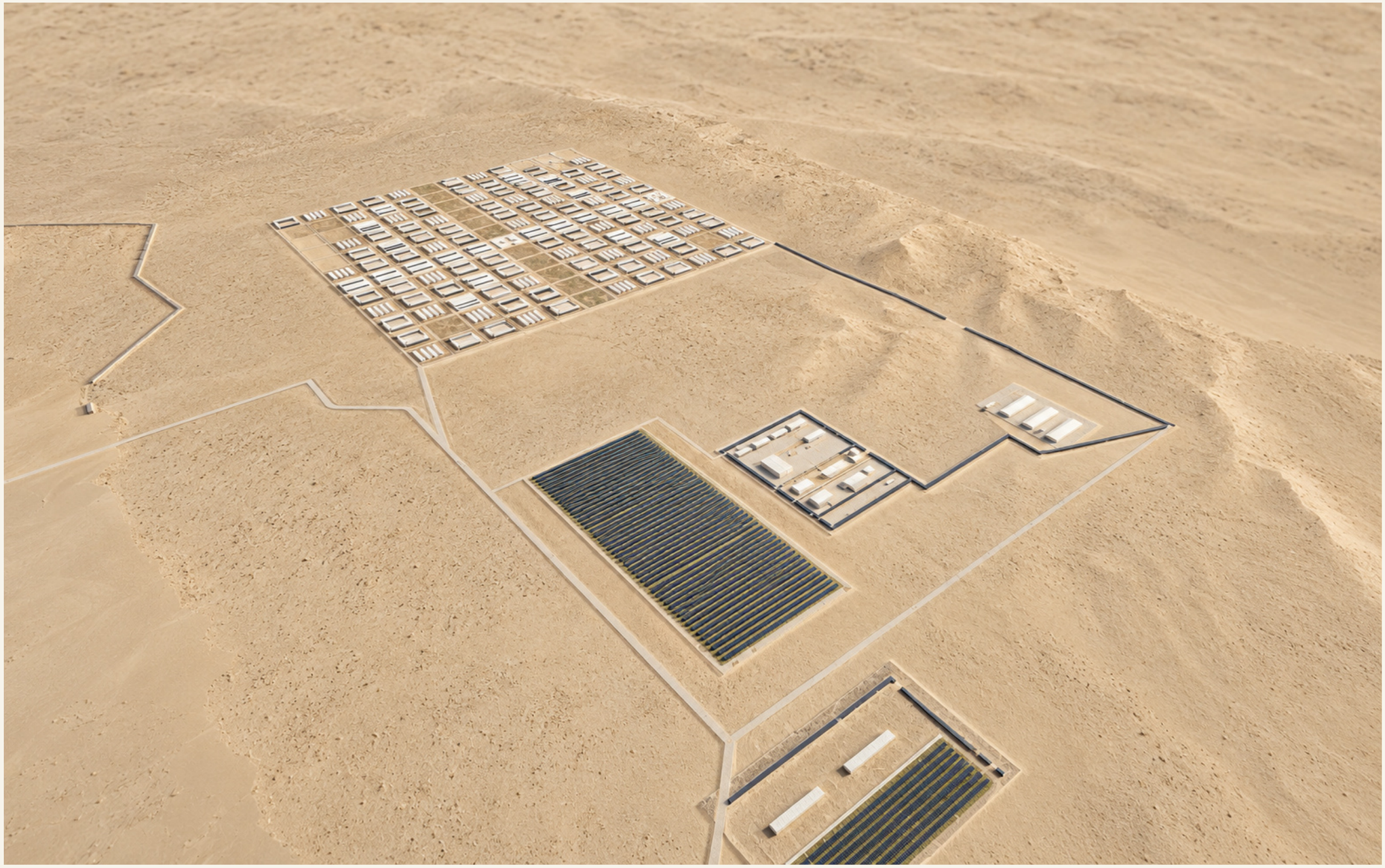
Four industrial data-centre halls at2 MW operating IT each are released 2 + 1 + 1. Repair, metered local drying trials and utilities have their own servicing. The adjacent 26.784 MWp PV scenario is conditional on module layout and grid study.

FARM / public learning and testing

Two farm-service sheds and a public shade promenade face eight new trial rows. Their small trial capacity is excluded from the main city PV account; crop bands have an explicit assumed water budget; commercial yields remain uncredited.

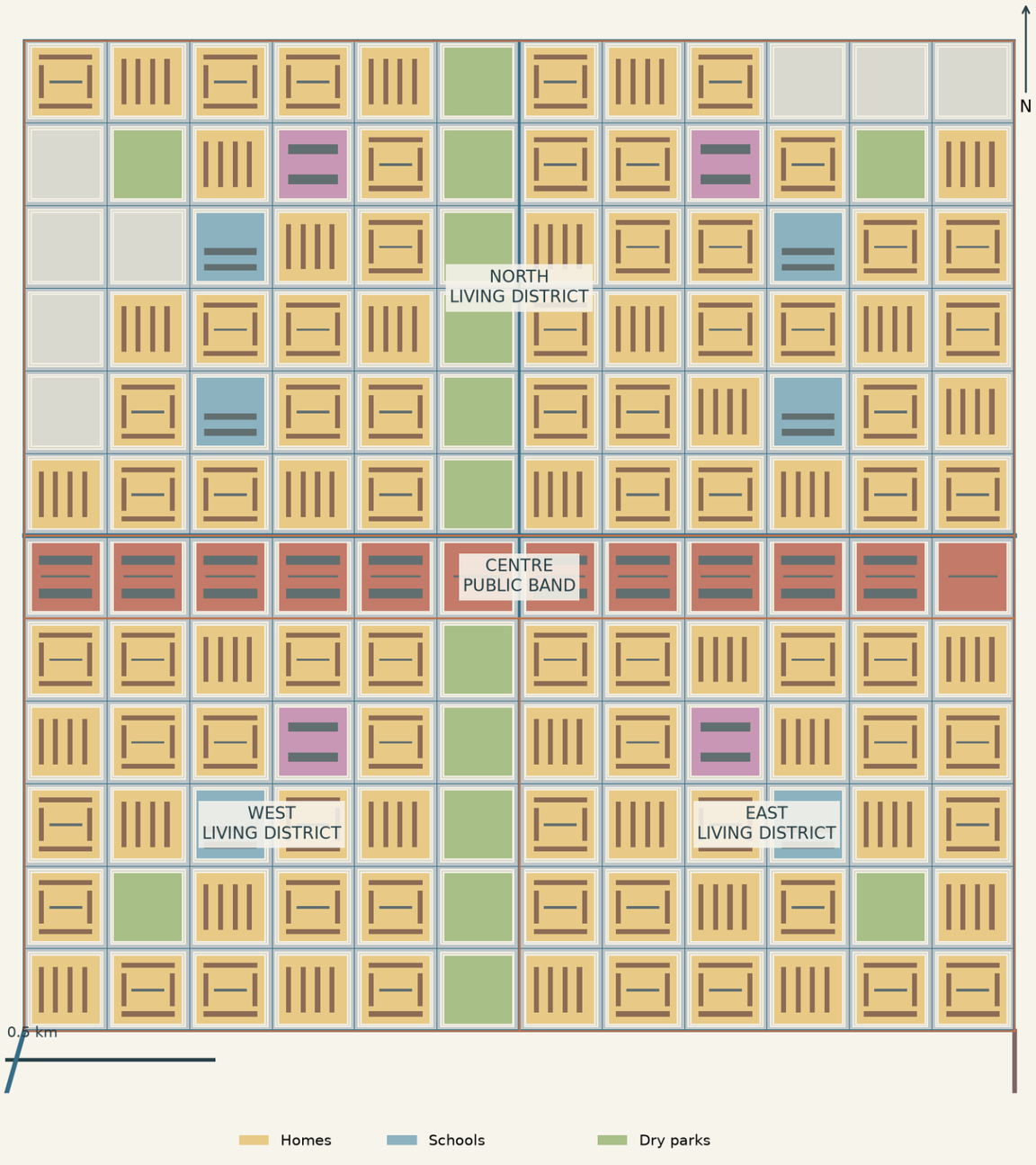
The town and its productive edges

Complete southwestern settlement / hypothetical full build-out / finished architectural-model concept; original camera control retained. This image does not depict the entire competition territory.



The town, independent productive campus, solar study and farm-learning edge retain their actual relative coordinates. Distant territorial districts remain in the full-site plan rather than being invented into the view.

Districts, blocks and the civic band

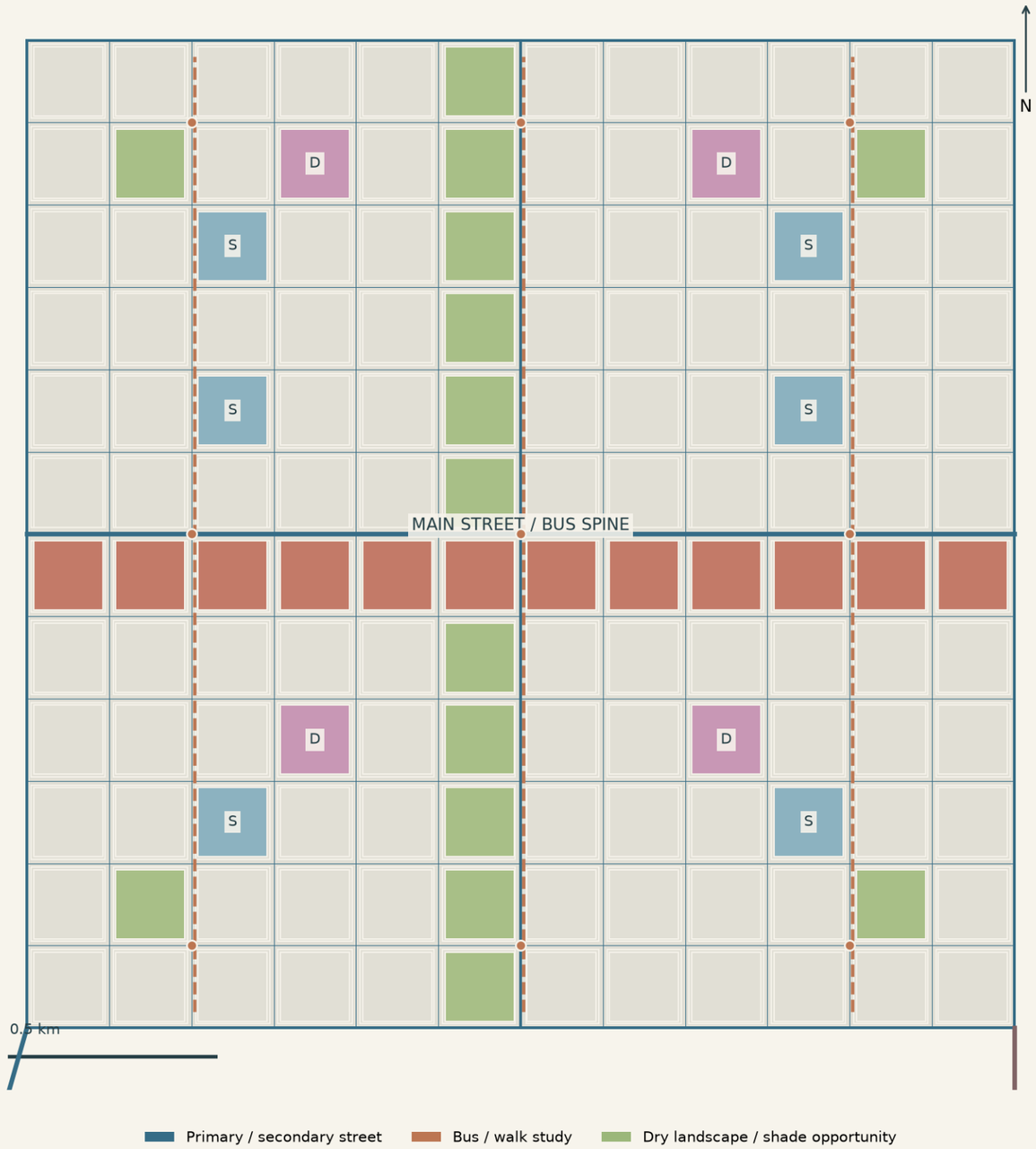


Three living districts.
The West is the first complete occupation area; the East extends the same service logic; the North completes the town.

Two public axes.
The east-west civic band holds shared destinations. The north-south dry commons gives a continuous public address.

Different housing grain.
Three-storey open court bars alternate with two-storey terrace lanes. The all-phase inventory yields 7,872 dwellings, a capacity ceiling rather than a proven market.

A hierarchy of streets and public movement

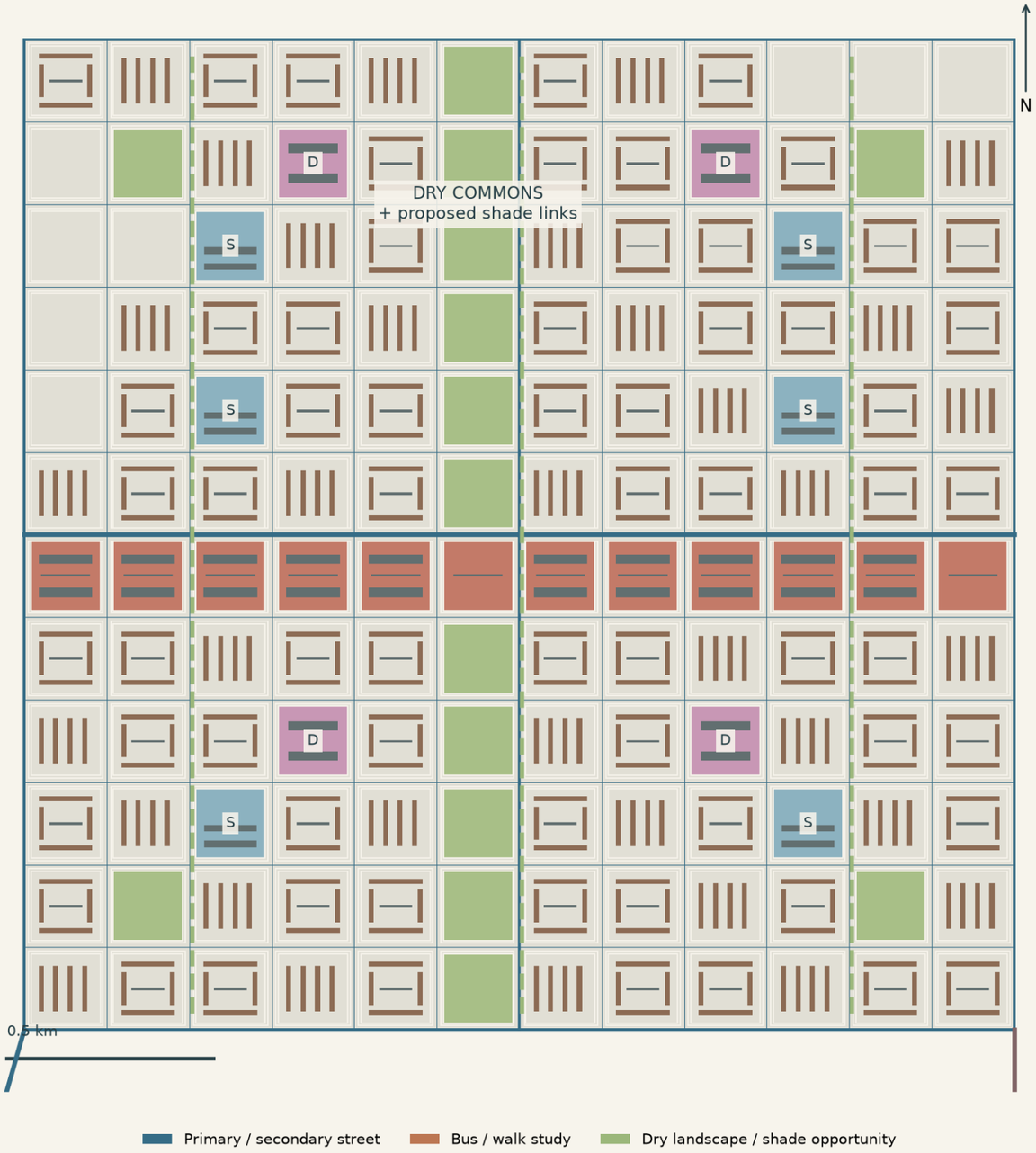


Blue: streets.
The 32 m civic Main Street and 28 m north-south avenue organise 20 m local streets. Widths are design reservations, pending engineering.

Rust: bus/walking study.
Stops mark a proposed network; frequency, accessibility and funding are not yet secured.

Outside the town.
A public southwest approach serves the farm and worker bus link. Production freight approaches on the east. Final junctions and emergency routes require transport review.

Dry commons and a shade network

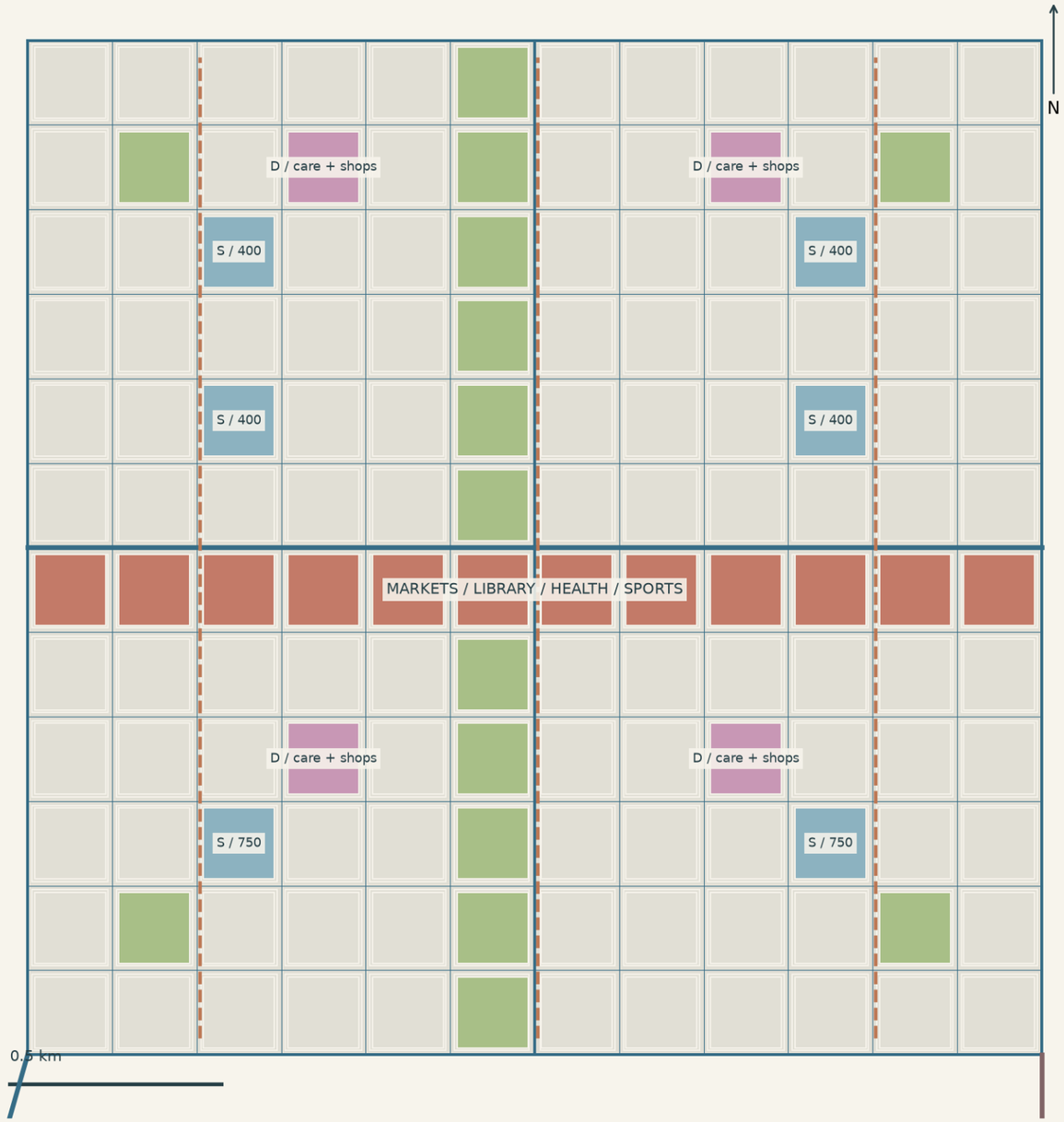


Landscape has different roles.
Dry parks and the commons provide public ground.
Source seasonal channels remain landscape/conveyance studies at territorial scale.

Green dashed links are proposals.
They identify future walking/shade opportunities on real street alignments, not established trees or guaranteed continuous shade.

Engineered shade first.
Courtyard, civic, worker and learning canopies are modelled explicitly. No irrigated lawn, decorative lake or mature planting capacity is credited.

Schools, care and daily destinations



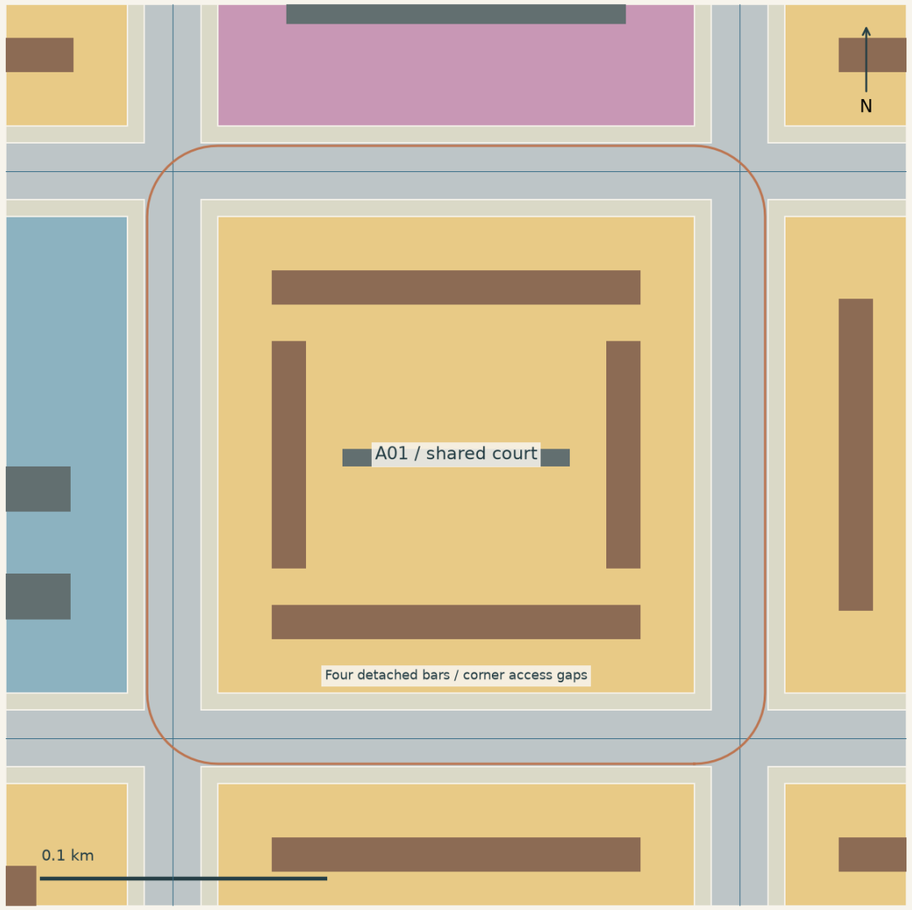
Services precede homes.
Two southern schools reserve 750 places each. Four northern schools reserve 400 each. The 3,100-place scenario exceeds an assumed 15% of 19,680 residents (2,952); demand, staffing and age mix need verification.

Four daily centres.
Care and neighbourhood shops supplement the larger markets, library, sports and health uses in the civic band.

No blanket five-minute claim.
Some trips across the 2.4 km town need the proposed bus and a measured safe-route audit.

A block is shared ground, not an employer compound

Two original housing types / build-out capacity from modelled buildings / detailed dwelling accessibility and tenure design remain next work.

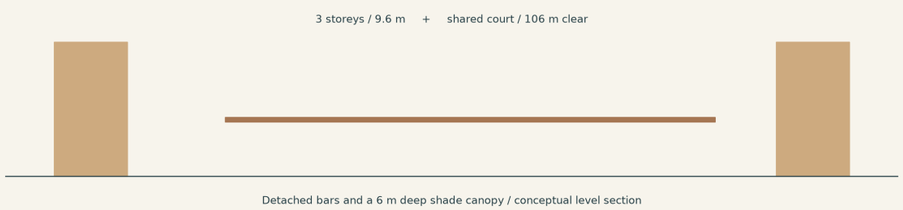


OPEN COURT BARS

Four separate bars define an open court. The two long bars each carry 30 dwellings; the two short bars each carry 18: 96 homes per court-block at three storeys. Corner gaps preserve public access; a 6 m deep canopy offers a shared address.

TERRACE LANES

Four two-storey rows each carry 12 dwellings: 48 homes per block. They offer a lower-rise counterpart, not a translated copy of the courtyard. Street frontage, emergency access and maintenance remain shared municipal obligations.

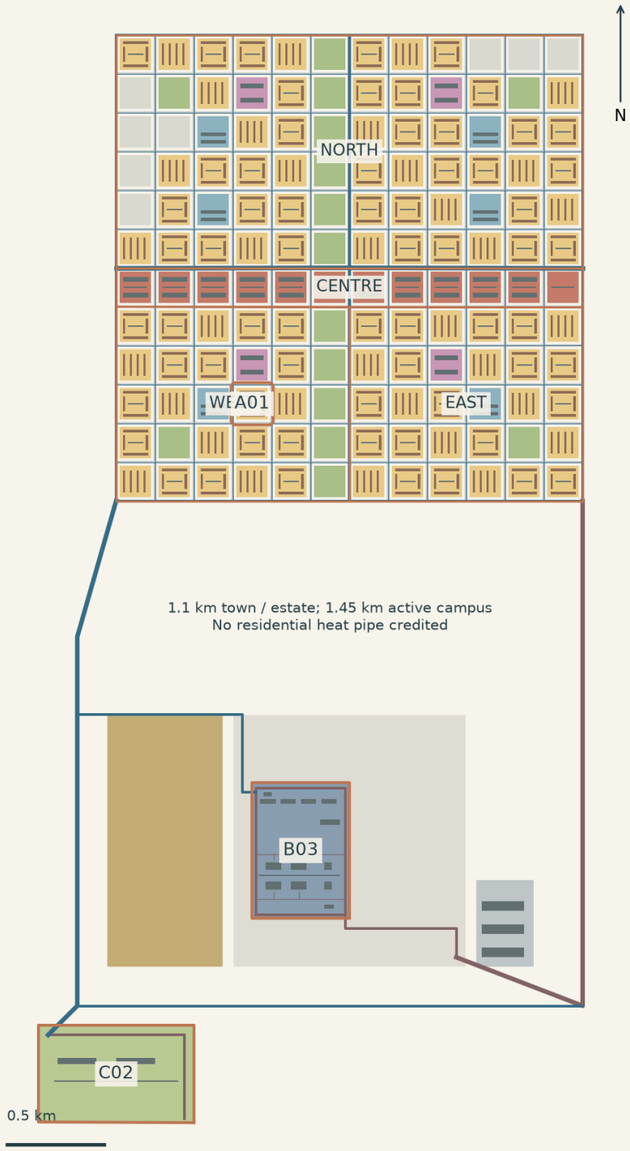


Detached bars and a 6 m deep shade canopy / conceptual level section

Both counts are declared design capacities, not sale plots, guaranteed household demand or surveyed building permissions. Landscape furniture and people in the finished images are illustrative.

Three places test three city promises

The nodes are located by the same coordinates as the master plan, not assembled into a fictional single view.



A01 / OPEN-COURT HOMES

West living district / P1. A complete residential court tests shared daily space, corner access and housing independent of an employer.

B03 / COMPUTE + REPAIR

Southern Works campus / P1-P3. Four staged modular groups test independent servicing and a limited local heat recipient. No claim that low-grade heat powers residential life.

C02 / FARM LEARNING EDGE

Southwest new trial campus / P1. Teaching, servicing and a public shade edge are arranged on new land outside the fixed pilots; the pilot interiors remain unchanged.

A01 / Open-court homes

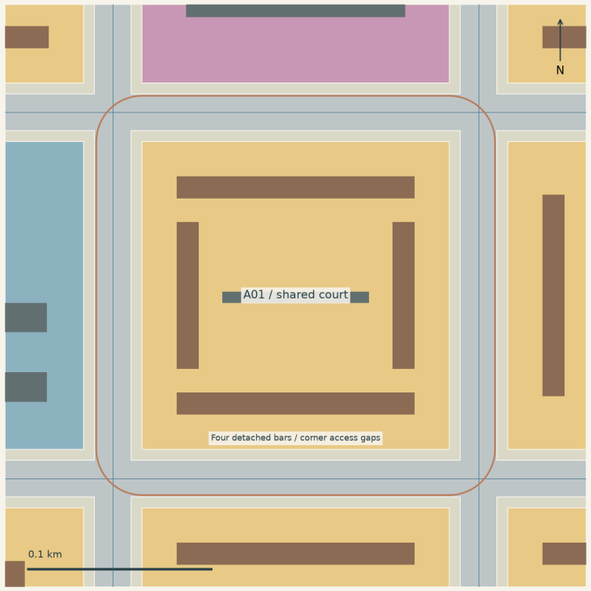
Three-storey housing, shared mineral court and an explicit shade band. Local aerial; full-build-out context.

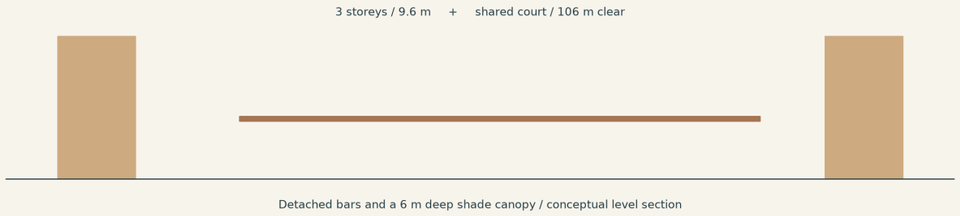


A01 / use, access and spatial detail

A separate geometry-derived eye-level camera, plan and conceptual section; image details are illustrative, not capacity evidence.







Detached bars and a 6 m deep shade canopy / conceptual level section

Residents enter through the open corners and meet under the existing shade structure. A housing cooperative or municipal operator must fund cleaning, accessible paving, lighting and canopy inspections. No cars are routed through the shared court.

B03 / An independent productive campus

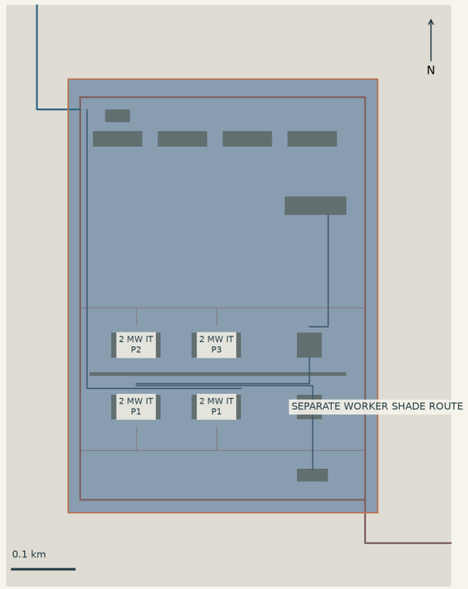
Four staged modular-service groups, repair sheds, utility servicing and a separate worker shade spine. Local aerial; all stages shown.



B03 / use, access and spatial detail

A separate geometry-derived eye-level camera, plan and conceptual section; image details are illustrative, not capacity evidence.





8 m service pavilions / 60 m service gap / 6 m worker shade

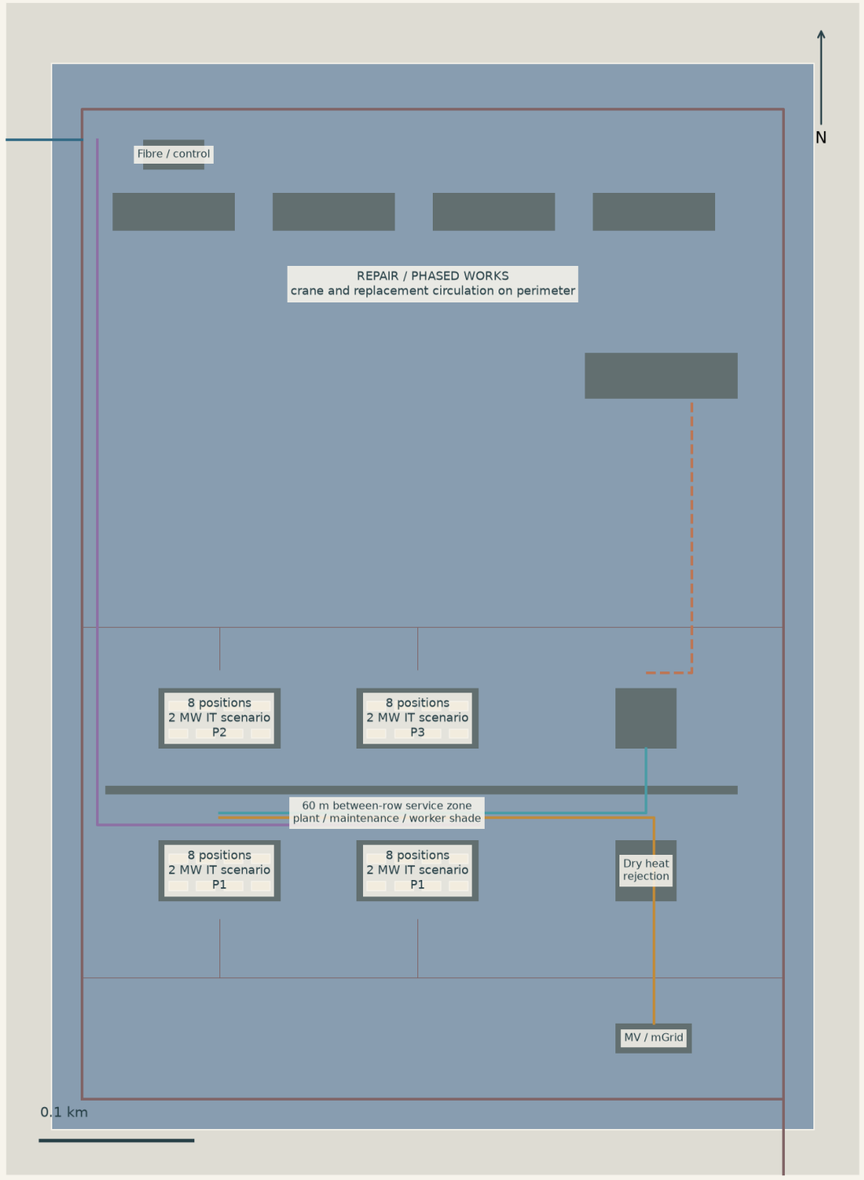


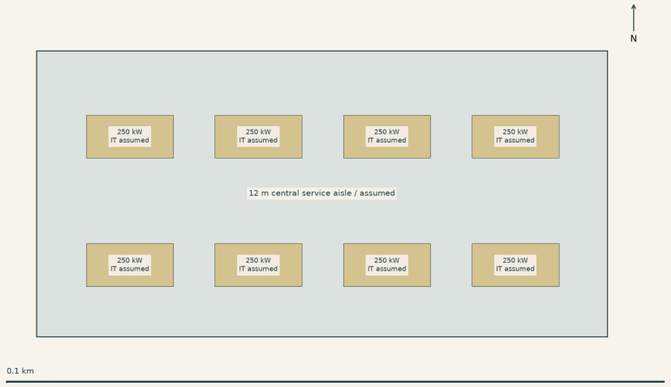
Covered modular positions; local heat test uncredited; no pipe to homes

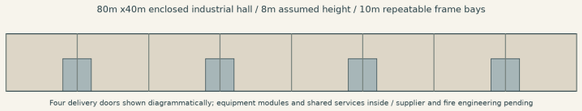
Workers reach the protected internal spine from the bus link. Freight takes the perimeter and service spurs. The registered human/machine interfaces are geometric flags, not proven-safe crossings. IT interruption contracts protect water/power limits; service budgets cannot depend on uninterrupted output.

B03 / compact modules and shared campus services

0.35 km2 serviced campus within1.56 km2 gross estate;1.21 km2 is unserviced future reserve with no capacity credit.







Four 80 x40m industrial halls each contain eight conceptual integrated equipment bays. Operating IT is assumed 250kW/position: P1/P2/P3 = 4/6/8 MW. Public XPOD reference: 6 racks x 56 kW =336 kW simple rack nameplate;1 MW power-block rating is not 1 MW IT.

50 m side gaps and 60 m between-row service space reserve delivery, replacement, shared plant/fibre/thermal routes and worker shade. Vendor layout, fire access and crane envelopes remain engineering checks. Heat rejection is provided; local drying earns no useful-heat credit yet.

C02 / A working farm-learning edge

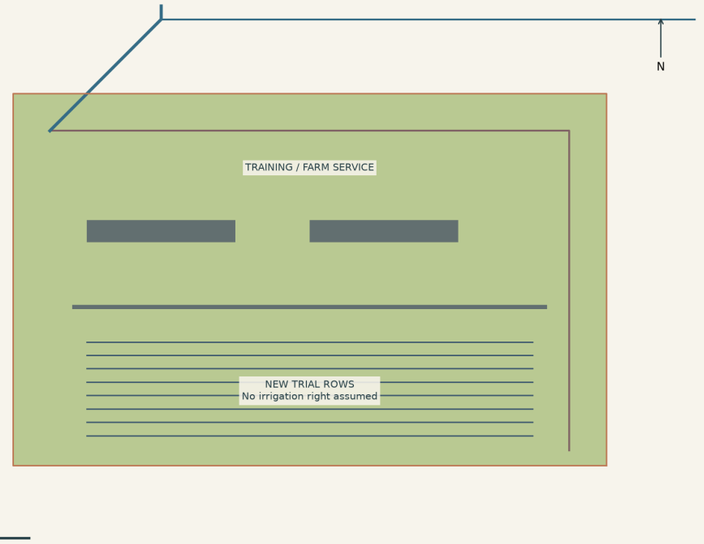
Two new service sheds, an accessible public promenade and eight trial PV rows. Outside fixed pilots; proposed forage and jojoba cultivation; irrigation entitlement unconfirmed.

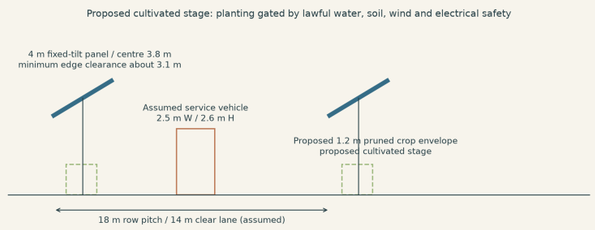


C02 / use, access and spatial detail

A separate geometry-derived eye-level camera, plan and conceptual section; image details are illustrative, not capacity evidence.







Visitors use the public promenade while farm technicians and deliveries use the rear/side service road. Forage and pruned jojoba occupy alternating declared bands below raised panels. Planting requires lawful water and soil/wind trials; visitors stay outside the working lanes.

C02 / agriculture under the photovoltaic rows

Production, maintenance and visitors have distinct ground. This is proposed cultivation, not an existing planted field or a redesign of the protected pilots.

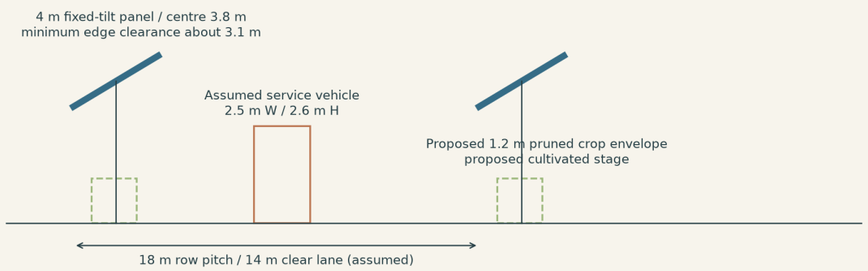


Two compatible crop systems.
Alternating 2 m bands carry forage and pruned jojoba. Four-metre panels have 3.8 m centre height and about 3.1 m minimum edge clearance; jojoba is held at 1.2 m. C02 has 18 m pitch, 14 m clear lanes and 100 m end headlands. All dimensions are assumed.

Wind and water.
Jojoba is wind-pollinated: provisional 1:8 male/female trial distribution along rows and both ends requires measured seasonal wind. Metered drip has separate valves and electrical isolation.

Real production budget.
New C02 plus main-field crop bands total 3.828 ha forage + 3.828 ha jojoba. Irrigation allowances 4000 /2000m3/ha/year give 22,968 m3/year. These are screening assumptions; lawful supply, soil and crop trials gate planting. No commercial yield credited.

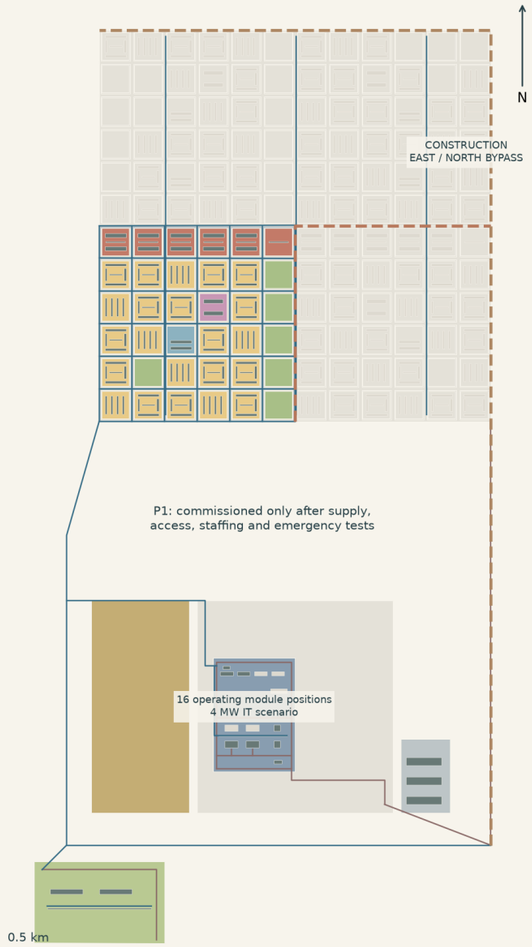
Proposed cultivated stage: planting gated by lawful water, soil, wind and electrical safety



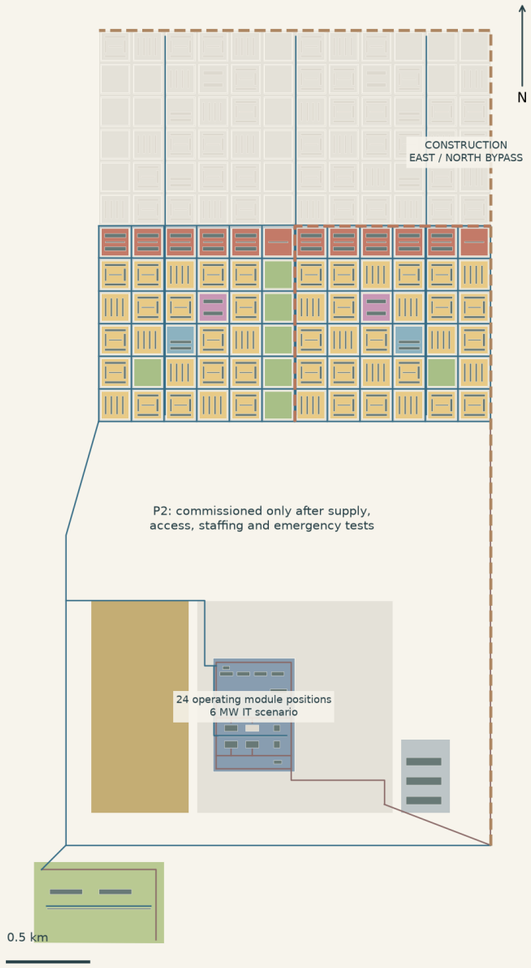
Commission operation, isolate the next works

Rust dashed routes reserve eastern/northern construction access; grey buildings remain unopened. These maps share the phase geometry and do not certify delivery rights.

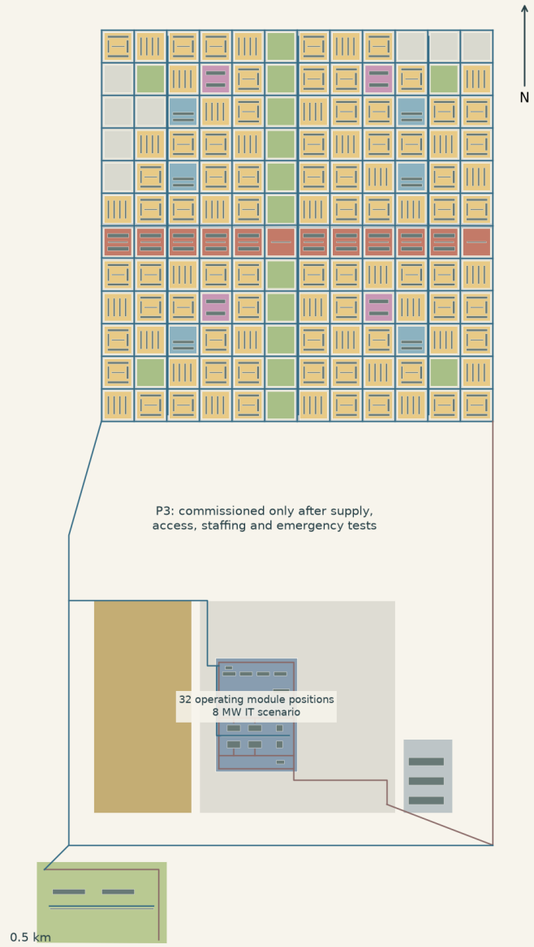
PHASE 1



PHASE 2



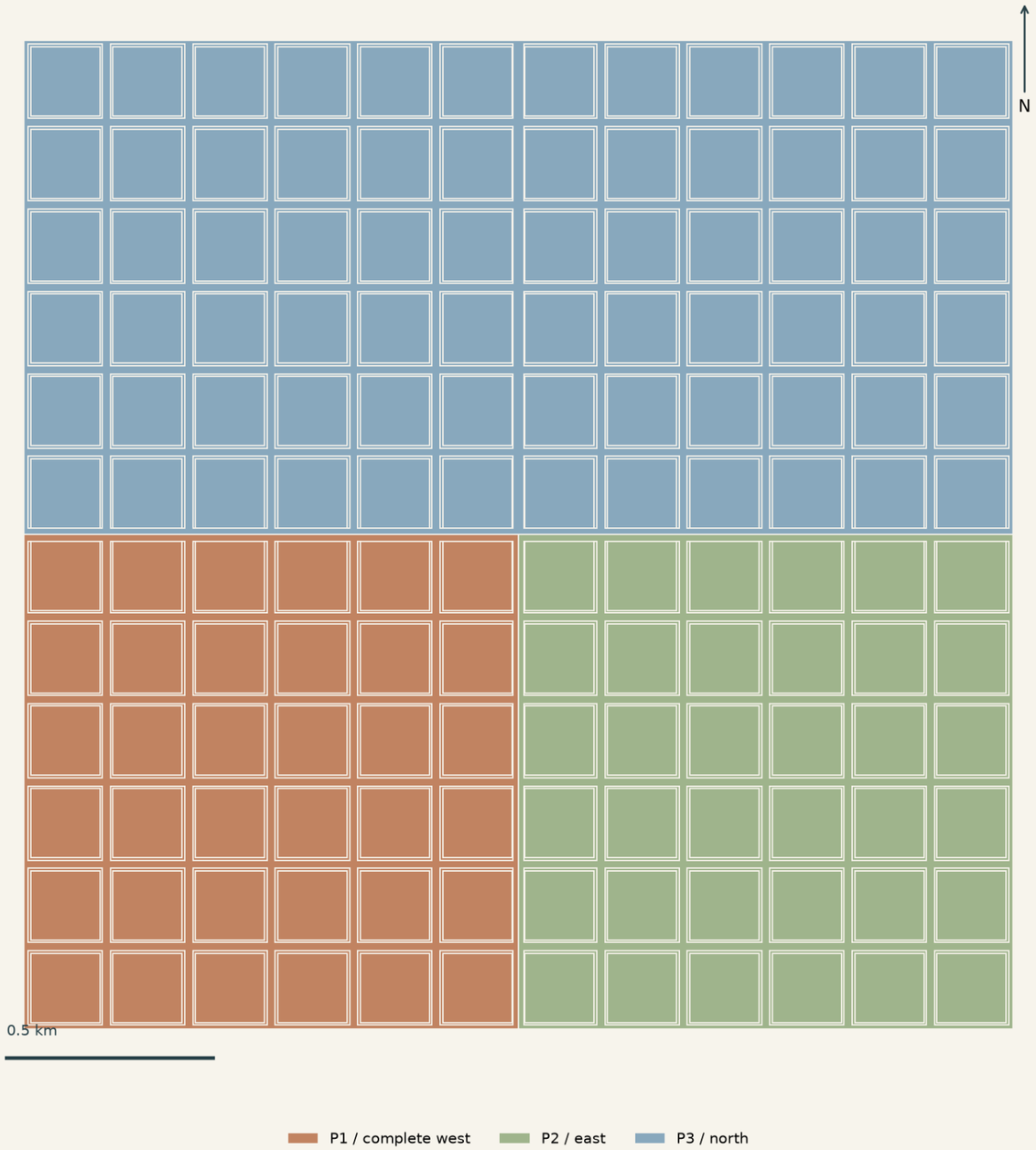
PHASE 3



Campus commissioning:16 /24 /32 positions at assumed 250kW operating IT each. Module delivery and essential-service testing precede energisation. Occupied West retains public access while future East/North works are fenced and bypassed.

Build a complete first phase; expand only when it works

Relative periods are scenarios. Phasing boundaries, homes, jobs and service obligations reconcile to the same building inventory.



P1 / Years 0-5 scenario

4,440 residents / 1,776 homes

Complete western neighbourhood, civic core, utilities and first productive halls. Funding and resource gates precede occupancy.

P2 / Years 5-10 conditional

9,720 residents / 3,888 homes

Eastern living district and additional productive capacity. Funding and resource gates precede occupancy.

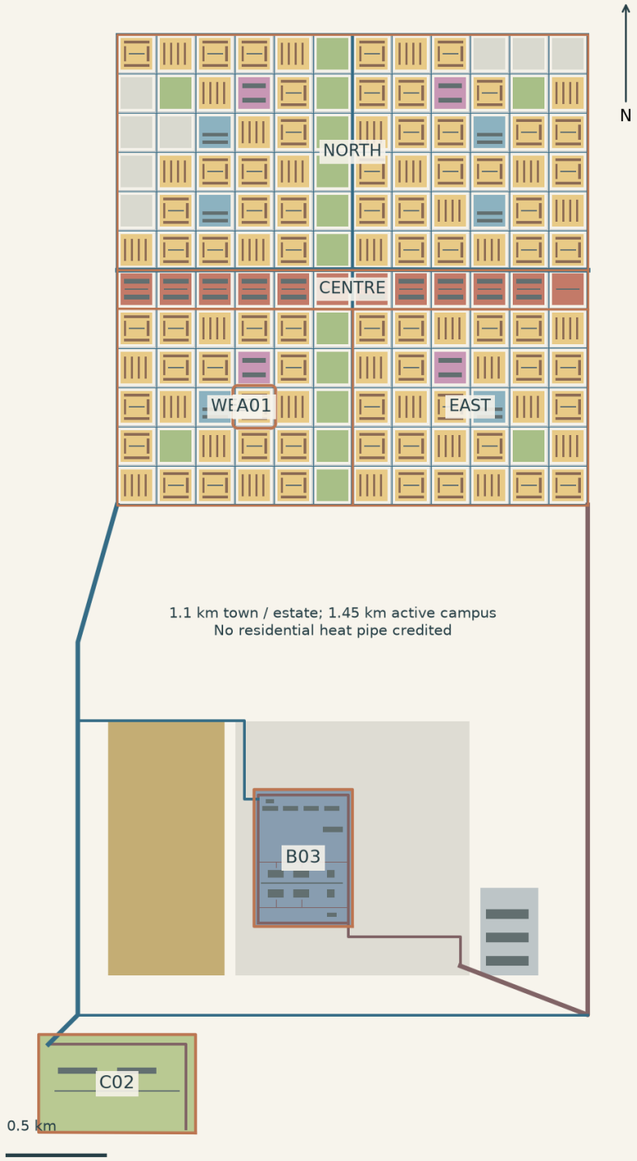
P3 / Years 10-20 conditional

19,680 residents / 7,872 homes

Northern neighbourhoods and final staged halls. Funding and resource gates precede occupancy.

The first neighbourhood must operate on its own

P1 scenario: 1,776 dwellings, 4,440 residents, 905 modelled jobs. These are separate capacities; no employment-demand balance is assumed.



Before the first keys

Lawful external access, water abstraction/supply, treatment, power and emergency access must be secured. The western school (750 places), care/shops and the western civic band open with occupied homes.

An operating budget, not a promise

Ring-fenced housing/public-service funding covers education, healthcare, street/shade maintenance and worker travel. Compute downtime cannot remove essential city services.

If later investment stops

The western neighbourhood retains its own school, daily care, markets, utility modules and access. Eastern/northern works are gated. The mismatch between 19,680 eventual residents and 2,150 modelled jobs remains a demand/employment verification need, not an assumed commuter solution.

Resource gates define what can be occupied

All figures below are declared scenarios derived from capacity; no water entitlement, grid contract, yield or climate measurement is supplied by these calculations.

WATER / FULL BUILD-OUT

774.0 thousand m3/y gross use

85 L/person/day domestic + 15 L/person/day civic, 35 L/job per workday +1,000m3/year panel cleaning, and an indicative 0.1 L/kWh compute makeup assumption. Reuse: 100,000 m3/y scenario. After 8% distribution loss, required fresh abstraction is 732,580 m3/y.

New forage/jojoba bands total7.656ha and include22,968m3/y assumed irrigation. Fixed pilot demand remains additional and unknown. No landscape irrigation or commercial crop yield is credited.

POWER / FULL BUILD-OUT

8 MW IT / four staged modular groups

P1/P2/P3 installed IT scenario: 4 / 6 / 8 MW. At assumed PUE 1.3, continuous full IT would use 91.1 GWh/y before town loads.

Explicit main PV rows reserve 26.784 MWp at an assumed 4 m panel width and 0.2 kWp/m2. At a synthetic 1,800 kWh/kWp/y yield, this is 48.2 GWh/y; it does not prove hourly matching or firm power.

Grid capacity, storage, weather and measured heat demand require independent study. No clean-energy percentage is claimed.

What this city chooses, and what it must still prove

The proposal trades shortest productive routes and maximum uninterrupted IT output for independent housing, public services and a clearer operating boundary.

A public-life decision

Living districts and the public band come first. Compute and production are arranged where service access can be controlled. This creates additional travel, utility cost and maintenance instead of claiming every adjacency is beneficial.

A capacity ceiling, not an instant new city

7,872 dwellings / 19,680 residents is the complete design capacity. Actual occupation requires housing demand, employment/livelihood evidence and independently funded services; the current 2,150 modelled jobs do not settle that question.

The next review has concrete questions

Survey and civil ground fit; lawful access and resource rights; noise, dust and emergency separation; an accessible walking/bus audit; crossing and canopy engineering; metered local drying demand; school staffing and annual operating budgets.

Traceable without overwhelming the presentation

Original controls, cameras, prompts, area/phase accounts and sources are in the separate technical appendix and machine files. Earlier V1 geometry is withdrawn from recommendation and retained for comparison. Neither polished images nor automatic file checks are independent design approval.